

A Memetic Path Planning Algorithm for Unmanned Air/Ground Vehicle Cooperative Detection Systems

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Abstract—The studies of Unmanned Air/Ground Vehicle (UAV/UGV) cooperative detection systems have received much attention due to their wide applications in the disaster rescue, target tracking, intelligent surveillance, and automatic package delivery missions. UAVs provide a broad view and have a fast speed in the air, while UGVs have sufficient load capacity and can serve as repeater stations on the ground. The path planning of a UAV/UGV cooperative system is an important but difficult issue, which aims to plan paths for both the UAVs and the UGVs in the system to cooperatively complete a mission. In this article, we consider the path planning problem of the UAV/UGV cooperative system for illegal urban building detection, by taking the limits of UGV speed, UAV load power, and UAV/UGV communication restriction into consideration. To solve this problem, we first model the path planning problem as a constraint optimization problem which tries to minimize an overall execution time for completing the illegal urban building detection tasks, and then propose a two-level memetic algorithm (called Two-MA) to solve the path planning problems of both the UAV and the UGV. Experiments on both synthetic and real-world data sets show the superiority of the proposed Two-MA over several state-of-the-art algorithms in solving the path planning problems of the UAV and UGV for illegal urban building detection tasks.

Note to Practitioners—This article was motivated by the task of detecting illegal buildings in cities by unmanned vehicles. Previous works mainly focus on path planning of either UAVs or UGVs in this task. This article proposes a new approach using an Unmanned Air/Ground Vehicle (UAV/UGV) cooperative system for detecting illegal buildings in parks, by taking the limits of UGV speed, UAV load power, and UAV/UGV communication restriction into consideration. This cooperative system consists of a UAV, UGV, and control center. The UAV equipped with cameras takes aerial photography in the air, and can transmit

collected photos to the control center. The UGV executes loading and transportation on the ground, and can serve as takeoff and landing platforms for the UAV. The control center executes computationally intensive tasks such as data transmission and processing, task scheduling, and vehicle coordination. To quickly complete all detection tasks, a memetic algorithm is proposed for path planning of both the UAV and the UGV. The simulated results show that the proposed algorithm enables the UAV/UGV cooperative system to visit all buildings in cities with a minimum task execution time.

Index Terms—Cooperative detection systems, illegal urban building detection, memetic algorithm, path planning, unmanned air/ground vehicle (UAV/UGV).

I. INTRODUCTION

THE Unmanned Air/Ground Vehicle (UAV/UGV) cooperative systems [2], [3] consist of UAVs, UGVs, and a control center, aiming to coordinate the UAVs and UGVs to complete real-world tasks under restrictions. For example, in the disaster relief tasks, UAVs obtain the geomorphic information and send the aerial map to the ground-monitoring center, while UGVs provide relief supplies to the disaster area based on the aerial map and the path planning generated by the ground-monitoring center [2]. In recent years, the studies of UAV/UGV cooperative detection systems have attracted much attention due to their wide applications in both the military and civilian fields, such as the intelligent surveillance and reconnaissance (ISR) missions [4]–[8], disaster relief [2], [3], [9]–[11], automatic package delivery [12]–[14], moving or static target tracking and detection [15]–[19], precision agriculture [20]–[22], geographic information collection [23], [24], mobile robot planning [25]–[27], robot obstacle avoidance [28], robot trajectory prediction [29], [30], emotion segmentation [31], and unmanned system control [32], [33].

UAVs in the air have a fast flight speed and broad vision, and therefore they can quickly bypass obstacles and traverse multiple locations. However, they are constrained by the flight distance due to limited power endurance [34]. UGVs on the ground have a good endurance capability and stability, and therefore they can load supplies and travel a long distance [35]. However, they are constrained by the limited vision and speed. A cooperation of UAVs and UGVs can comprehensively combine their advantages over individuals in completing complicated tasks in terms of speed, vision, capability, and stability. For example, in illegal building/park

Manuscript received 5 January 2021; accepted 15 February 2021. Date of publication 10 March 2021; date of current version 13 October 2022. This article was recommended for publication by Associate Editor C. Yang and Editor H. Gao upon evaluation of the reviewers' comments. This work was supported in part by the National Nature Science Foundation of China under Grant U1713212, Grant 61836005, Grant 62073225, Grant 61803269, and Grant 61976142; in part by the National Key Research and Development Program of China under Grant 2020YFA0908700; in part by the Natural Science Foundation of Guangdong Province-Outstanding Youth Program under Grant 2019B151502018; in part by the Natural Science Foundation of Guangdong Province under Grant 2020A1515010790; in part by the Technology Research Project of Shenzhen City under Grant JSGG20180507182904693 and Grant JCYJ20190808174801673; and in part by the Public Technology Platform of Shenzhen City under Grant GGF2018021118145859. (Corresponding author: Lijia Ma.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TASE.2021.3061870>.

Digital Object Identifier 10.1109/TASE.2021.3061870

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detection scenario, buildings/parks are widely distributed over cities and they usually change over time. In this case, it is difficult for individuals to manually collect these building information at all time. For UAV/UGV cooperative systems, a UGV can carry and charge the UAV while the UAV can take photos for buildings. This UAV/UGV cooperation can effectively complete tasks of path planning for illegal urban building/park detection.

Path planning is a common but important issue in UAV/UGV cooperative systems for performing real-world complicated tasks [17], [36]–[40]. Actually, the works on path planning have received much attention, and have successfully applied to many real-world applications [41]–[43]. Unlike previous works, path planning of UAV/UGV cooperative systems needs to consider the limits of UAV endurance capacity, UGV load capacity, obstacles, and UAV/UGV communication restrictions. Generally, this path planning problem is first modeled as an optimization problem with an objective that evaluates the total path length of UAVs or UGVs, and then some heuristic or metaheuristic algorithms are devised to solve this optimization problem.

In recent years, many works have been proposed for the path planning of UAVs and UGVs for performing real-world complicated tasks. For examples, in the moving tracking task, Tanner [17] planned paths for a UAV/UGV cooperative system, in which UGVs are allocated into specific cells in an area while UAVs scan these cells based on the planned paths. In the target-tracking mission, Yu *et al.* [15] modeled the path planning as a probability maximization problem, which uses a dynamic occupancy grid to represent moving target states. In the reconnaissance mission, Manyam *et al.* [44] developed a mixed integer linear programming (MILP) algorithm with a branch-and-cut strategy to plan paths of the UAV and UGV, by taking the communication constraints between the aerial and ground vehicles into consideration. They further transformed the path planning problem of UGVs and UAVs into a MILP problem, and used pruning strategies to speed up operations [45]. Gao *et al.* used sensor fusion and situational assessments to path planning for autonomous UGVs [46], [47]. In the autonomous package delivery tasks, Mathew *et al.* [12] solved a shortest path planning problem, in which trucks can only drive on streets while quadrotors should visit all given locations to deliver packages. In the large-scale mapping applications, Maini and Sujit [48] proposed a greedy strategy to plan paths for both UGVs and UAVs, in which UAVs much stay around UGVs. It is noted that these afore-mentioned works mainly focus on path planning of either UAVs or UGVs, and meanwhile they neglect the area clustering property of tasks [49]–[51]. Accordingly, these works are restricted to the air or ground task applications, and they cannot work well for large-scale problems. Actually, path planning of UAV/UGV cooperative systems can be well generalized to the path length-based traveling salesman problem (TSP) [52], and it can be solved by some classic heuristic algorithms, such as the ant colony [53], simulated annealing [54], genetic optimization [55], branch-and-cut optimization [45], and memetic optimization algorithms [56]–[58].

In this article, we study the path planning of a UAV/UGV cooperative system that is equipped with cameras for illegal building detection in parks and cities. The UAV has advantages in the speed and vision, while the UGV has advantages in the load capability and stability. However, the UAV is generally limited by its energy and load capability, while the UGV is limited by its speed and vision. In a city, the supervised buildings are widely distributed over a large area, and the potential unapproved building components are located in different heights. Although some works have been proposed for the path planning of either UAV or UGV detection systems, the path planning of UAV/UGV cooperative systems remains unsolved. Moreover, traditional UAV or UGV detection systems are difficult to detect nonstandard buildings of cities as they neglect the limits of UGV speed, UAV load power, and UAV/UGV communication restriction [48], [59].

To detect nonstandard buildings in cities, in this article, we propose a UAV/UGV cooperative detection system, by taking the limits of UGV speed, UAV load power, and UAV/UGV communication restriction into consideration. In this system, the UGV can carry the UAV and meanwhile charge the drone automatically, while the UAV can take a photo for the nonstandard buildings from an aerial perspective. Moreover, we first model the detection of urban illegal problem as a time-aware TSP optimization problem, and then propose a two-level memetic path planning algorithm (called Two-MA) to solve this problem. Two-MA first adopts a K-means algorithm to divide the buildings of cities into clusters, in which buildings in the same clusters are closer than those in different clusters. Then, Two-MA executes a hierarchical path planning, in which a cluster-level memetic algorithm is used for path planning of the UGV for visiting buildings across different clusters while a node-level memetic algorithm for path planning for the UAV in each cluster. The main contributions of this article are summarized as follows.

- 1) We propose a novel UAV/UGV cooperative detection system for executing nonstandard building detection tasks in urban areas, which comprehensively considers the advantages and limits of the UAV and UGV.
- 2) We model the path planning problem of detection of nonstandard buildings as a time-aware TSP optimization problem, and propose the Two-MA algorithm to solve this problem.
- 3) Based on the UAV/UGV cooperative system, we design a memetic algorithm to plan the path for the cooperative UAV and UGV in the system, with the objective to minimize the total task execution runtime.
- 4) Extensive experiments on both synthetic and real-world data sets show the superiority of the proposed Two-MA over the state-of-the-art algorithms for the detection of nonstandard buildings in cities.

This work is an extension of our previous work [1] which also studies the path planning of the UAV/UGV cooperative systems for the detection of nonstandard buildings in cities. Unlike our previous work, we consider the limits of UGV speed and UAV load power in this work. Moreover, we take the power consuming and charging of UAV into consideration

TABLE I
NOTATIONS OF GRAPHS AND RULE DIVISIONS OF THE SYSTEM MODEL

Notations in graphs	Rule divisions
Nodes	Buildings
Edges	Possible paths between buildings
Edge weighs in airs	Minimum path length between buildings in airs
Edge weighs in grounds	Minimum path length between buildings in grounds
Cluster	A set of buildings that are close to each other

during the task execution. In addition, we model the detection of urban illegal problem as a time-aware TSP optimization problem, rather than a path-aware TSP optimization problem. More specifically, this work tries to minimize the task execution time of the UAV/UGV cooperative systems, while our previous work aims to minimize the visiting path of the UAV/UGV cooperative systems. Finally, this work designs a more strictly coupled cooperative way between the UAV and the UGV than our previous work. More specifically, this work allows the UGV to charge the UAV and considers the waiting time of the UGV.

The rest of this article is organized as follows. Section II introduces some related backgrounds. Section III illustrates our UAV/UGV cooperative system and describes the proposed memetic path planning algorithm. Section IV analyzes experimental results on both synthetic and real-world data sets. Section V gives conclusions and some possible future works.

II. SYSTEM MODEL AND PROBLEM FORMULATION

In this part, the system model is first introduced, and then the problem formulation is given. Table I gives some notations of graphs and rule divisions of the system model.

A. UAV/UGV Cooperative System for Illegal Building Detection

Generally, a UAV/UGV cooperative system consists of UAVs, UGVs, and a control center. UAVs equipped with cameras perform aerial photography in the air, and transmit the collected data to the control center for further analysis. UGVs execute loading and transportation on the ground, and serve as the takeoff and landing platforms for the UAVs. The control center executes the compute-intensive tasks such as data transmission and processing, task scheduling, and vehicles coordination [12], [20]. A simulated UAV/UGV cooperative system is shown in Fig. 1, in which the UGV serves as the takeoff and landing platform for the UAV, and the control and coordination center is replaced by a laptop. This UAV/UGV cooperative system is widely used for performing real-world complicated tasks. For example, in the intelligence surveillance and reconnaissance (ISR) mission, the control center first plans paths for the UAV and UGV, and then processes data transmitted from the UAV/UGV to give further instructions. A comprehensive survey for the UAV/UGV cooperative system is shown in [60].

In this study, we try to use a UAV/UGV cooperative system to detect illegal buildings in a city while considering the limits of UGV speed, UAV load power, and UAV/UGV



Fig. 1. Simulated UAV/UGV cooperative system.

communication restriction. In reality, the buildings of a city are widely distributed into a large area with different heights, and they usually change over time. Moreover, the UAV in the air has a fast flight speed and broad vision, which can quickly negotiate obstacles and traverse multiple locations. However, it is constrained by the flight distance due to the limited power endurance. Furthermore, it has a limited communication distance and needs to communicate with the UGV or control center all the time. The UGV on the ground has a good endurance capability and stability, which can load supplies and travel a long distance. However, it is constrained by the limited vision and speed. Therefore, the single UAV or UGV system is hard to complete the tasks of detecting illegal buildings in a city. The cooperation of the UAV and UGV can comprehensively combine their advantages such as fast speed, long transportation distance, taking photo and data mining. Therefore, compared with either UAV or UGV systems, the UAV/UGV cooperative system can be more applicable to tasks of detecting illegal buildings in a city.

Our UAV/UGV cooperative system is composed of a UAV, a UGV, and a control center. The UAV is responsible for taking photos for each building, while the control center is used to detect whether the building is illegal or not based on the photos from the UAV. The UGV provides the load services for the UAV and control center, and provides charging services for the UAV. Potential applications of UAV/UGV cooperative system include sensor data collection, air ground cooperative monitoring, package delivery, etc. For examples, in the sensor data collection, sensors are widely distributed over the monitoring area. In this case, a clustering method can be first used to transform sensor monitoring networks into two-level tasks, and then the UGV and UAV are used to monitor the ground and air networks, respectively. In the package delivery, the UGV can first carry the UAV to the package's designated areas, and then use the UAV to assign the packages to users around the areas.

We let $G = \{\mathcal{V}, \mathcal{E}, D_a, D_g\}$ be the related information of detected buildings over a city, where $\mathcal{V}$ denotes buildings while $\mathcal{E}$ represents possible roads between buildings. Moreover, D_a (or D_g) represents the shortest distances between buildings in the air (or ground). More specifically, each element D_a^{ij} of D_a represents a minimum distance between buildings i and j in the air. We let s_a and s_g denote the speeds of the UAV and UGV, respectively, and let d_m represent a maximum

communication distance between the UAV and other devices (like the UGV and control center). Moreover, we assume that the power consumption (or charging) pattern of the UAV follows a linear model $f(t) = u_f \cdot t$ (or $g(t) = u_g \cdot t$), where u_f (or u_g) is a constant control parameter and $f(t)$ (or $g(t)$) is the consumption (or charging) power in the time period t .

B. Problem Formulation in the UAV/UGV Cooperative System for Detection of Illegal Buildings

For the illegal urban building/park detection scenario, buildings are widely distributed over cities and they usually change over time. The UAV/UGV cooperative detection systems can be well used to replace individuals to perform detection tasks at regular intervals.

Given the information of buildings $G = \{\mathcal{V}, \mathcal{E}, D_a, D_g\}$, the speed s_a of the UAV, the speed s_g of the UGV, and the power consumption (charging) pattern $f(t) = u_f \cdot t$ (or $g(t) = u_g \cdot t$), the detection of illegal buildings aims to find an optimal path planning for the UAV and UGV, so as to take a minimum of task execution time to take photos for all buildings under the constraints that the UAV needs to communicate with the UGV or control center all the time and the UAV has a limited power endurance. We let $P = \{v_{p_1}, v_{p_2}, \dots, v_{p_n}, v_{p_1}\}$ denote a possible path started and ended from v_{p_1} , where $v_{p_i} \in \mathcal{V}$, $i = 1, 2, \dots, n$, is the i th detecting building while n is the number of buildings. Our path planning problem for the UAV/UGV cooperative detection system can be modeled as the following constrained time-aware TSP problem:

$$TSP(P) = \min T_a(P) + T_g(P) + T_a^w(P) \quad (1)$$

where $T_a(P)$ is the flying time of the UAV on the path, while $T_g(P)$ is the driving time of the UGV on the path. Moreover, $T_a^w(P)$ is the waiting time that is used for the UAV to charge the power on the UGV. $T_a(P)$ ($T_g(P)$) can be computed by the total flying (driving) distance of the UAV (UGV) across the path divided by the speed of the UAV (UGV), while $T_a^w(P)$ can be computed based on the power consumption (or charging) pattern $f(t) = u_f \cdot t$ (or $g(t) = u_g \cdot t$). Formally, $T_a(P)$ is computed as follows:

$$T_a(P) = \sum_{i=1}^{n-1} \frac{D_a^{p_i p_{i+1}}}{s_a} \cdot C(p_i, p_{i+1}) \quad (2)$$

where $C(p_i, p_{i+1}) \in \{0, 1\}$ denotes whether the transportation from node v_{p_i} to $v_{p_{i+1}}$ is visited by the UGV or UAV. Specifically, $C(p_i, p_{i+1}) = 1$ indicates that the transportation from node v_{p_i} to $v_{p_{i+1}}$ is visited by the UAV, while $C(p_i, p_{i+1}) = 0$ means the completion of one cluster (relatively dense area within the city), and the UGV is on its way to the next cluster.

$T_g(P)$ is computed as follows:

$$T_g(P) = \sum_{i=1}^{n-1} \frac{D_g^{p_i p_{i+1}}}{s_g} \cdot (1 - C(p_i, p_{i+1})) + \frac{D_g^{p_n p_1}}{s_g}. \quad (3)$$

$T_a^w(P)$ is evaluated as follows:

$$T_a^w(P) = (T_a(P) - T_a^C) \cdot \frac{u_f}{u_g} - \left(T_g(P) - \frac{D_g^{p_n p_1}}{s_g} \right) \quad (4)$$

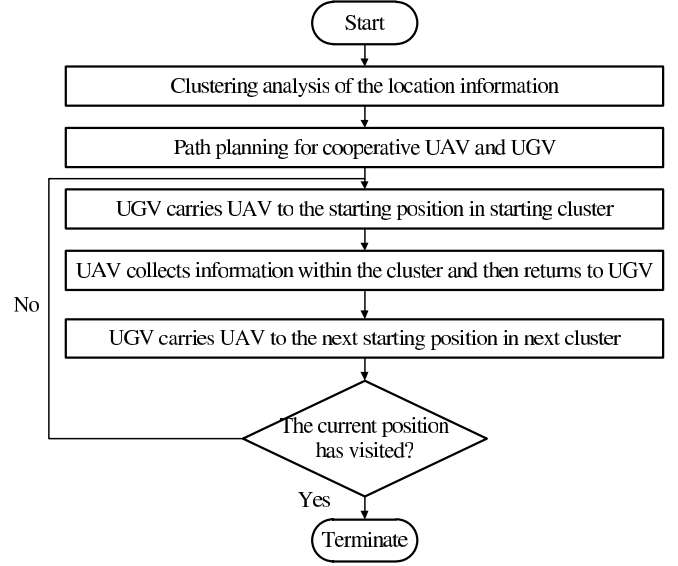


Fig. 2. Framework of the cooperative model of the UAV/UGV system.

where T_a^C is the flying time endurance of the UAV. Recall that u_f (or u_g) is the constant control parameter in the linear power consumption (or charging model). $(T_a(P) - T_a^C) \cdot u_f / u_g$ evaluates the charging time of the UAV for completing all tasks, while $(T_g(P) - D_g^{p_n p_1} / s_g)$ computes the possible charging time on the UGV.

The constraint of (1) is that the UAV needs to stay in the nearby of the UGV for communication at all times.

It is noted that this constrained time-aware TSP problem is an NP-hard problem, and classical path planning algorithms for UAVs or UGVs are hard to solve this problem as they neglect the constraints of the problem and the cooperations between the UAVs and UGVs.

III. OUR SOLUTION

To solve the time-aware TSP problem in (1), in this part, we first propose a novel cooperative model for the UAV/UGV cooperative system, and then propose a memetic path planning algorithm.

The whole framework of the cooperative model is shown in Fig. 2. In this model, the buildings in cities are first divided into a set of clusters, in which buildings in the same cluster are close whereas those in different clusters are always far from each other. Then, the UGV is scheduled to take the UAV and control center to traverse different clusters, while the UAV is used to take photos for buildings in the clusters. Due to the communication constraint, the distance between the UAV and the UGV is always less than a maximum communication distance d_m . This would limit the size of clusters.

In the following, the main steps of the cooperative model are introduced, including the clustering analysis and the path planning for the UAV and UGV.

Step 1) (Clustering Analysis): Generally, the buildings in a city are widely distributed over a large area, and their nonstandard objects are located in different heights. To quickly plan paths for all buildings, we first divide the areas with

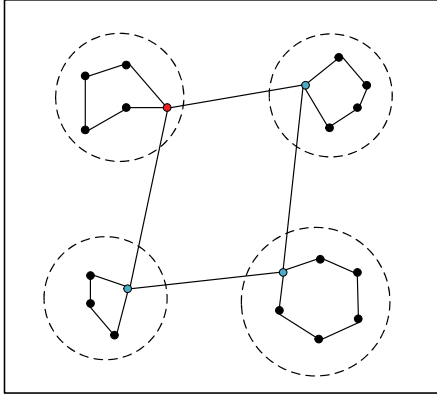


Fig. 3. Graphical representation of the system model. The dotted circles represent clusters, while the red and blue nodes represent the start positions of the UAV/UGV system and the UAV path in clusters, respectively.

buildings into a set of clusters, in which buildings in the same cluster are close whereas those in different clusters are far to each other. Actually, many clustering algorithms can be well used to divide buildings over cities into clusters, such as K-means clustering [61], mean-shift clustering [62], density-based spatial clustering [63], and hierarchical clustering [64]. Here, we choose the K-means clustering algorithm [61] to generate building clusters based on the global position system (GPS) data. This is because K-means can achieve a good tradeoff between the clustering efficiency and time consumption.

Considering the limits of UAV endurance capacity and UGV load capacity, we use the UAV to gather image information of buildings in the clusters and use the UGV to carry the UAV to traverse different clusters. Moreover, to ensure communications between the UAV and the UGV, we set the diameter of all clusters less than d_m .

Step 2) (Path Planning for the UAV): In a cluster, the UAV first takes off from the UGV, and then traverses all buildings within the cluster. Finally, it returns to the UGV for power charging. To reduce the power consumption of the UAV and the total operation time of the UAV/UGV cooperative system, the control center needs to find the shortest path for the UAV to travel all nodes in a cluster, which can be modeled as a typical TSP problem in the node level. As the speed of the UAV is known *a priori*, the TSP problem of each cluster can be considered as a time-aware TSP problem, which tries to minimize the flying time of the UAV to traverse all buildings.

Step 3) (Path Planning for the UGV): After the UAV collects all image information of buildings within a cluster, the UGV carries the UAV to the next cluster. The UGV loaded with UAV traverses across clusters, which can be modeled as a typical time-aware TSP in the cluster level. In the transportation process across different clusters, the UAV is charged by the UGV. The charging time of UAV is decided by the path length across two different clusters. If the path length is too short, the drone's power supply of the UAV is not enough to allow it to traverse the next cluster, and the whole system needs to wait for charging the UAV. To minimize the total runtime

Algorithm 1 K-Means Clustering Algorithm

```

1: Input: Number of clusters:  $k$ ; Maximum number of iterations:  $q_{max}$ ; Position data of urban buildings:  $\mathcal{V}$ ;
2:  $k$  buildings are randomly selected as centroids, and each centroid  $j$  is set as an initial cluster  $\mathcal{C}_j$ ,  $j = 1, 2, \dots, k$ .
3: Set  $q = 1$ .
4: while  $q < q_{max}$  do
5:   for each building  $i \in \mathcal{V}$  do
6:     Building  $i \in \mathcal{V}$  is divided into the cluster  $\mathcal{C}_j$  that has the minimum distance with building  $i$ .
7:   end for
8:   for each cluster  $\mathcal{C}_j$  do
9:     Computes the averaged building position of all buildings in the cluster  $\mathcal{C}_j$ , and set this building position as the centroid of  $\mathcal{C}_j$ .
10:  end for
11:   $q \leftarrow q + 1$ .
12: end while
13: Output: cluster divisions  $\mathcal{C} = \{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_k\}$ .

```

of the whole cooperative system, we also use the proposed memetic algorithm for the path planning of the UGV among different clusters. Here, the objective of the path planning is to minimize the total runtime of the system, including the transportation and waiting time of the UGV.

Using the two-level (node-level and cluster-level) memetic algorithms, Two-MA can effectively solve the optimization problem in (1).

Based on the planned paths for the UAV and UGV, the heterogeneous cooperative system collects the geographic information of the buildings as follows. First, the UGV carries the UAV to the initial position of a cluster. Then, the UAV collects the information of the buildings within the cluster based on the UAV path. Next, the UGV carries the UAV to the next cluster based on the UGV path. The steps above are executed until all information of the buildings are collected. An illustration representation of the operation framework of the cooperative system is given in Fig. 3.

In the following, the K-means clustering algorithm and the proposed memetic algorithm are described.

A. K-Means Clustering Algorithm

The urban building position data, collected by the GPS, contain two dimensions that represent the longitude and latitude of buildings. We let $\mathcal{V}$ denote the position data of urban buildings. The K-means clustering algorithm is used to divide the urban building into k clusters, where k is the predefined number of clusters. The framework of the K-means clustering algorithm is given as **Algorithm 1**.

As known from **Algorithm 1**, the K-means clustering algorithm first finds k initial centroids and sets them as initial clusters (see line 2). Then, it iteratively allocates all buildings into the k clusters that have the minimum distances with these buildings (see lines 5–7), and updates the centroids of clusters (see lines 8–10). It is terminated when the predefined maximum number of iterations q_{max} is reached (see line 4).

Algorithm 2 Framework of Two-MA

```

1: Input: Cluster divisions:  $\mathcal{C} = \{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_k\}$ ; Starting
   points of all clusters:  $\mathcal{V}_s$ .
2:  $P_g \leftarrow \mathbf{MA}(\mathcal{V}_s)$ .
3: for each cluster  $\mathcal{C}_j$  do
4:    $P_a^j \leftarrow \mathbf{MA}(P_g, \mathcal{C}_j)$ .
5: end for
6:  $P \leftarrow \{P_g, P_a^1, P_a^2, \dots, P_a^k\}$ .
7: Compute  $TSP(P)$  based on (1)–(4).
8: Output:  $TSP(P)$  and  $P$ .

```

Algorithm 3 MA()

```

1: Input: Population size:  $S_{pop}$ ; Maximum number of gener-
   ations:  $G_{max}$ ; Crossover probability:  $p_c$ ; Mutation proba-
   bility:  $p_m$ ; The size of mating pool:  $N_m$ ; the tournament
   size:  $N_t$ ;
2:  $Pop = \text{Initialization}(S_{pop})$ ;
3: repeat
4:    $Pop_{parent} \leftarrow \text{Selection}(Pop, N_m, N_t)$ ;
5:    $Pop_{child} \leftarrow \text{GeneticOperation}(Pop_{parent}, p_c, p_m)$ ;
6:    $Pop_{new} \leftarrow \text{LocalSearch}(Pop_{child})$ ;
7:    $Pop \leftarrow \text{UpdatePopulation}(Pop_{new})$ ;
8: until The current iteration reaches  $G_{max}$ .
9: Output: Convert the fittest individual in  $Pop$  into the path.

```

Algorithm 4 Initialization()

```

1: Input: Population size:  $S_{pop}$ .
2: for  $i = 1$  to  $S_{pop}$  do
3:    $P_a^i$  (or  $P_g^i$ )  $\leftarrow$  Generate a random sequence with size  $n$ ,
     where  $n$  is the number of visited buildings.
4: end for
5: Output: Population  $Pop = \{P_a^i : i = 1, 2, \dots, S_{pop}\}$ .

```

B. Two-MA for Path Planning for the UAV/UGV

Two-MA aims to minimize the objective in (1) for the UAV and UGV by simultaneously optimizing a population of solutions. Two-MA first uses the cluster-level memetic algorithm for path planning across clusters for the UGV, and then uses the node-level memetic algorithm for path planning for the UAV in each cluster. Here, the cluster-level and node-level memetic algorithms have the same operations, and they work on different levels of objects. We use the function “MA()” to denote both the cluster-level and node-level memetic algorithms. The framework of Two-MA for path planning is given as **Algorithm 2**.

MA() consists of a genetic algorithm and a local search. The genetic algorithms uses the initialization, selection, genetic operation, and population update to explore solutions, while the local search uses a greedy strategy to exploit good solutions. **Algorithm 3** gives the framework of MA(), and the corresponding operations of MA() are detailed next.

1) Encoding and Initialization:

a) Encoding: We assume that there are n buildings that need to be visited by the UAV (or UGV). Then, for each

Algorithm 5 Selection()

```

1: Input: Size of mating pool:  $N_m$ ; Tournament size:  $N_t$ ;
2: for  $i = 1$  to  $N_m$  do
3:   Randomly choose  $N_t$  solutions from  $Pop$ .
4:   Choose the solution in the chosen solutions with the
     highest fitness values, and put this solution into  $Pop_{parent}$ .
5: end for
6: Output: Parent population  $Pop_{parent}$ .

```

Algorithm 6 GeneticOperation()

```

1: Input: Parent population:  $Pop_{parent}$ ; Crossover probabili-
   ty:  $p_c$ ; Mutation probability:  $p_m$ .
2: for each pair of solutions in  $Pop_{parent}$  do
3:   if (A randomly generated value  $r \leq p_c$ ) then
4:     Generate two random number  $r_1$  and  $r_2$ .
5:     Generate two offsprings solutions by swapping the  $r_1$ 
     to  $r_2$  genes' segment of two parent solutions.
6:     Process duplicate genes of two offsprings solutions,
     and then put them into  $Pop_{child}$ .
7:   end if
8: end for
9: for each solution in  $Pop_{child}$  do
10:  if (A randomly generated value  $r \leq p_m$ ) then
11:    Generate two random number  $r_1$  and  $r_2$ .
12:    Generate a mutated solution by swapping the  $r_1$ th
     gene and  $r_2$ th gene of the original solution, and use it to
     replace the original solution in  $Pop_{child}$ .
13:  end if
14: end for
15: Output: Child population:  $Pop_{child}$ .

```

individual, a sequence with n values is randomly generated. We let $P_a = \{P_a^1, P_a^2, \dots, P_a^n\}$ (or P_g) denote the solution encoding for the UAV (or UGV). The gene value P_a^i denotes the building label, while the gene position i denotes the visiting sequence. For example, the randomly generate sequence is (3, 1, 5, 9, 6, 2, 7, 8, 4, 10). This means that there are 10 buildings that need to be visited, and the visiting sequence is 3, 1, 5, 9, 6, 2, 7, 8, 4, 10. After visiting building 10, the UAV (or UGV) returns to the building labeled as “3”.

b) Initialization: In the initialization step, the Initialization() function (see **Algorithm 4**) randomly generates S_{pop} sequences as the population Pop for UAV (or UGV). More specifically, for each individual, a sequence with size n is first randomly generated, and then its fitness value is computed by (1)–(4).

2) Selection: The aim of Selection() (see **Algorithm 5**) is to preserve individuals with a high fitness to the next generation. In this research, the typical tournament selection strategy is adopted, which randomly selects N_t chromosomes from the initial population Pop and preserves the best one into the mating pool for N_m times. The tournament selection strategy guarantees the diversity and superiority of the population.

3) Genetic Operation: The GeneticOperation() (see **Algorithm 6**) contains the crossover and mutation operations

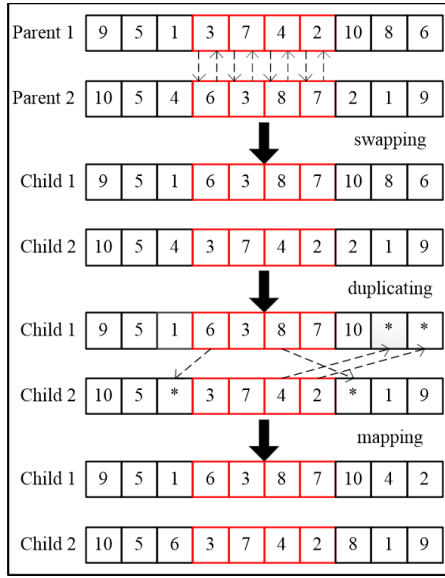


Fig. 4. Schematic illustration of the crossover operation. Here, the generated values r_1 and r_2 are 4 and 7, respectively.

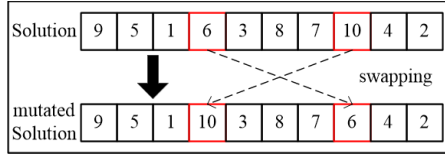


Fig. 5. Schematic illustration of the mutation operation. Here, the generated values r_1 and r_2 are 4 and 8, respectively.

to guide MAs for exploration. More specifically, crossover creates offspring solutions by recombining and inheriting the genes of parent solutions, while mutation encourages genetic diversity by tuning the solutions' genes.

Partial-mapped crossover is widely used for genetic operation to solve TSP problems due to its simplicity and good inheritance of genes from parent solutions. Given two parent solutions, two random numbers r_1 and r_2 are first generated, and then crossover operation swaps the selected part of two parent solutions' genes from r_1 to r_2 (see lines 4–5). The crossover operation may introduce duplicate genes to the original solution. In this case, we conserve the swapped genes of children solutions, and map the duplicate genes in other segments to the corresponding genes of conflicting ones in the swapped segments (see line 6). Fig. 4 shows a detailed crossover of two parent solutions.

Swap mutation is used as mutation operation due to its simplicity and randomness. For each solution in the population, two random numbers r_1 and r_2 are first generated, and then mutation operation swaps the r_1 th gene and r_2 th gene in the solution (see lines 11–12). A schematic illustration of the mutation operation is shown in Fig. 5.

4) *Local Search*: The LocalSearch() (see **Algorithm 7**) is essential for exploitation and convergence. For the time-aware TSP optimization problem, the refinement of a solution needs to optimize the path time by combining solution's genes. To improve the performance of the genetic algorithm, the evolution reverse operation is introduced into the local search [57]. After genetic operation, local search is carried out on each

Algorithm 7 LocalSearch()

```

1: Input: Child population:  $Pop_{child}$ .
2: for each solution in  $Pop_{child}$  do
3:   Generate two random number  $r_1$  and  $r_2$ .
4:   Generate a new solution by reversing the genes from  $r_1$ 
     to  $r_2$  of the original solution.
5:   if (new solution's fitness  $\geq$  original fitness) then
6:     Put the new solution into  $Pop_{new}$ .
7:   else
8:     Put the original solution into  $Pop_{new}$ .
9:   end if
10: end for
11: Output: New population:  $Pop_{new}$ .

```

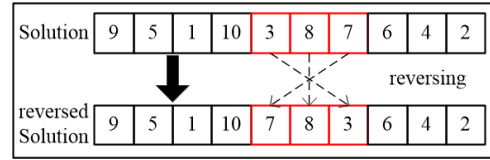


Fig. 6. Schematic illustration of the evolution reverse operation. Given two randomly generated integers $r_1 = 5$ and $r_2 = 7$, we reverse the corresponding segment of the original chromosome when this reversion can improve the fitness of the chromosome.

solution in the population. As is shown in Fig. 6, two random numbers r_1 and r_2 are first generated, and then the selected genes from r_1 to r_2 of a solution are reversed (see lines 3–4). Next, new individual's fitness need to be recomputed. If the generated individuals have a higher fitness than the original ones, then this reversion is accepted. Otherwise, the original individuals are retained for the next iterations (see lines 5–8).

5) *Update Population*: In this step, we choose a population of individuals with high fitness values (low-task execution time) from the current and offspring populations as the new population for the next generation. This step guarantees the evolution of the population at each iteration.

6) *Computational Complexity of Two-MA*: As known from Algorithm 2, Two-MA executes the function MA() $k+1$ times, where k is the number of building clusters in cities. Moreover, as known from Algorithm 3, each MA() needs to execute the initialization, selection, genetic operation, and local search. More specifically, the initialization needs to be executed one time, which has a computational complexity $S_{pop} \cdot n/k$, where S_{pop} is the size of population while n/k denotes the averaged number of buildings in a cluster. Moreover, the selection and genetic operation need to be executed G_{max} times, each of which has a computational complexity $N_t \cdot n/k$, where G_{max} is the maximum number of iterations and N_t is the size of offsprings. In addition, the local search needs to be executed G_{max} times, each of which has a computational complexity $N_t \cdot n/k$. Generally, $S_{pop} \leq G_{max} \cdot N_t$.

Therefore, our Two-MA optimization has an overall computational complexity $G_{max} \cdot N_t \cdot n$.

IV. EXPERIMENTAL RESULTS

In this section, we test the proposed memetic algorithm (Two-MA) on both synthetic and real data sets to show its effectiveness. Moreover, we compare **Two-MA** with several

TABLE II
SETTINGS OF PARAMETERS OF ALL EVOLUTIONARY ALGORITHMS

Parameters	Meaning	Values
G_{max}	The number of iterations	200
S_{pop}	Population size	100
N_m	Size of the mating pool	$\frac{S_{pop}}{2}$
N_t	Tournament size	5
P_c	Crossover probability	0.9
P_m	Mutation probability	0.05

TABLE III
SETTINGS OF THE COOPERATIVE SYSTEM

Systems	Key parameters	Specific values
UAV (A)	Velocity	50km/h
	Endurance	60min
	Charging time	180min
UAV (B)	Velocity	50km/h
	Endurance	30min
	Charging time	80min
UAV (C)	Velocity	60km/h
	Endurance	38min
	Charging time	90min
UGV	Velocity	30 km/h
UAV-UGV	Communication Distance	10 km

TABLE IV
CONFIGURATION OF OUR SYNTHETIC DATA SETS

Area Index	Number of points	Number of clusters
1	200	10
2	400	20
3	600	30
4	800	40

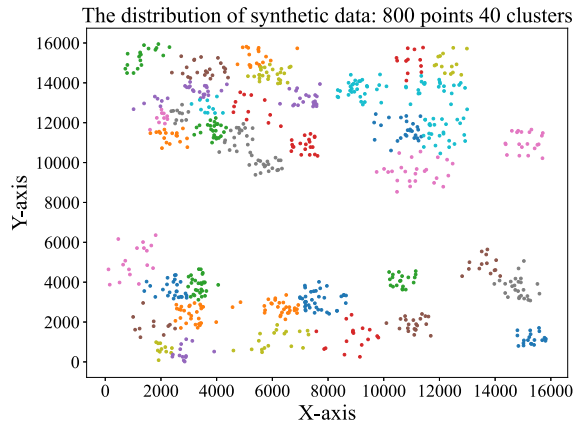


Fig. 7. Representation of the synthetic data in a coordinate axis. Here, points with the same colors are in the same clusters.

classical TSP algorithms, including the genetic algorithm (Two-GA) [55], ant colony optimization (Two-ACO) [65], and greedy algorithm (Two-Greedy) [65]. All results are averaged over 30 independent trials under parameters setting in Table II and system setting in Table III. The three types of UAVs in Table III are DJI M600 pro (A), DJI Phantom 4 pro (B), and DJI M200 (C), while the UGV Autolabor is used as a take-off, landing, and charging platform for the UAV.

A. Experimental Settings

1) *Data Sets*: In this experiment, we use four synthetic and real data sets to validate the effectiveness of our algorithm.

TABLE V
CONFIGURATION OF OUR REAL-WORLD DATA SETS

Area Index	City districts	Points and Clusters
1	Nanshan	92 points, 6 clusters
2	Nanshan, Futian	150 points, 10 clusters
3	Baoan, Guangming	284 points, 20 clusters
4	Baoan, Guangming Longhua, Nanshan	441 points, 30 clusters

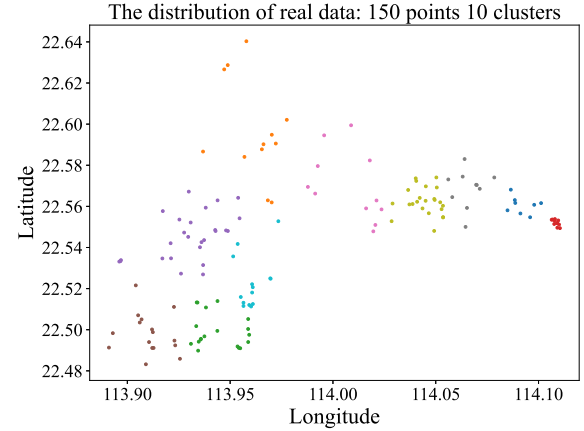


Fig. 8. Distributions of the realistic data. The solid dots with different colors represent different clusters. Here, there are 150 parks and 10 clusters in Nanshan and Futian District.

a) *Synthetic data sets*: They consist of four different sizes of planar point sets. The specific configuration of these synthetic data sets is shown in Table IV. We use python to generate planar point sets in a given range. More specifically, for the data set with 800 points and 40 clusters, we first randomly generate 40 cluster centers, and then randomly generate 19 adjacent points for each cluster. The radius between points is randomly selected from 1.6, 1.2, 0.8, and 0.6 km. To better illustrate those data, we plot the distribution of a synthetic data with 800 points in Fig. 7.

b) *Real-world data sets*: We derive the coordinates of all parks in Shenzhen City, and the longitude and latitude of these parks are collected from Baidu Map. In these data, the parks are divided into four groups based on the districts of the city. Table V gives some basic information about the real-world data sets, and Fig. 8 visually shows the distribution of parks in Nanshan and Futian District.

From Table V, we can see that there are four real-world data sets with different scales. From Figs. 7 and 8, we can see that both the synthetic and real-world data sets have heterogeneous clusters in sizes and scales, and the buildings of these data are widely distributed over Shenzhen city.

2) *Criteria*: To evaluate the performance of the proposed model and algorithm, the total path distance and task execution time of the UAV/UGV cooperative system are adopted as criteria. The total path distance includes the path distance of the UAV and UGV, while the task execution time includes the running time of UAV, the running time of UGV, and the waiting time of UGV for charging UAV to complete all tasks of detecting illegal buildings in a city.

3) *Comparison Algorithms*: Greedy algorithm (Two-Greedy): it adopts a greedy strategy with a minimum path

TABLE VI
RESULTS OF ALL ALGORITHMS ON THE SYNTHETIC DATA SETS

Setting	Data index	Algorithms	Distance (km)			Time (hour)			
			UAV	UGV	Total	UAV	UGV	UGV-wait	Total
A	1	Two-GA	81.01	29.86	110.9	1.620	1.979	0.9834	3.599
		Two-MA	71.73	38.32	110.1	1.435	1.329	0.05184	2.764
		Two-ACO	71.37	29.81	101.2	1.427	1.437	0.4429	2.864
		Two-Greedy	81.73	34.84	116.6	1.635	2.125	0.9638	3.760
	2	Two-GA	150.8	50.63	201.4	3.016	6.152	4.465	9.168
		Two-MA	133.3	120.0	253.3	2.666	5.021	1.021	7.686
		Two-ACO	134.0	46.90	180.9	2.681	5.203	3.639	7.884
		Two-Greedy	154.8	52.73	207.6	3.097	6.542	4.784	9.639
	3	Two-GA	238.2	86.27	324.5	4.764	11.44	8.567	16.21
		Two-MA	212.2	242.7	454.9	4.245	9.761	1.671	14.01
		Two-ACO	211.3	72.72	284.0	4.226	9.835	7.410	14.06
		Two-Greedy	244.4	76.30	320.7	4.888	11.95	9.403	16.83
	4	Two-GA	269.4	118.9	388.2	5.387	13.30	9.340	18.69
		Two-MA	238.4	320.0	558.4	4.768	11.33	0.6609	16.09
		Two-ACO	238.7	91.95	330.6	4.774	11.53	8.464	16.30
		Two-Greedy	275.4	94.67	370.1	5.509	13.89	10.74	19.40
B	1	Two-GA	81.38	29.86	111.2	1.628	3.115	2.120	4.743
		Two-MA	71.60	46.28	117.9	1.432	2.507	0.9646	3.939
		Two-ACO	71.34	29.67	101.0	1.427	2.635	1.646	4.062
		Two-Greedy	82.41	33.68	116.1	1.648	3.288	2.165	4.936
	2	Two-GA	150.5	50.02	200.5	3.009	6.788	5.120	9.797
		Two-MA	133.0	116.2	249.2	2.661	5.786	1.913	8.447
		Two-ACO	133.4	47.37	180.8	2.668	5.923	4.344	8.591
		Two-Greedy	153.8	53.27	207.1	3.077	7.168	5.392	10.24
	3	Two-GA	237.2	84.62	321.8	4.745	11.43	8.609	16.17
		Two-MA	212.4	227.8	440.2	4.248	10.02	2.429	14.27
		Two-ACO	211.6	72.32	283.9	4.231	10.11	7.695	14.34
		Two-Greedy	245.6	78.01	323.6	4.912	12.07	9.473	16.99
	4	Two-GA	267.7	116.9	384.5	5.353	13.08	9.185	18.43
		Two-MA	237.0	309.8	546.9	4.740	11.34	1.007	16.08
		Two-ACO	237.0	91.36	328.3	4.740	11.47	8.425	16.21
		Two-Greedy	273.6	94.30	367.9	5.472	13.62	10.47	19.09
C	1	Two-GA	82.11	29.61	111.7	1.368	1.829	0.8421	3.197
		Two-MA	71.63	38.41	110.0	1.194	1.353	0.07234	2.546
		Two-ACO	71.46	29.59	101.1	1.191	1.460	0.4736	2.651
		Two-Greedy	82.42	34.73	117.1	1.374	2.009	0.8512	3.382
	2	Two-GA	151.0	48.69	199.6	2.516	4.565	2.942	7.081
		Two-MA	133.6	107.8	241.4	2.226	3.799	0.2058	6.025
		Two-ACO	133.8	46.31	180.1	2.230	3.912	2.368	6.142
		Two-Greedy	155.2	53.11	208.3	2.587	4.908	3.137	7.495
	3	Two-GA	236.9	85.70	322.6	3.949	7.969	5.113	11.92
		Two-MA	211.1	196.5	407.6	3.518	6.863	0.3140	10.38
		Two-ACO	210.8	70.74	281.5	3.513	6.966	4.608	10.48
		Two-Greedy	244.2	76.41	320.6	4.070	8.434	5.888	12.50
	4	Two-GA	269.7	117.6	387.3	4.496	9.332	5.412	13.83
		Two-MA	238.1	229.4	467.4	3.968	7.931	0.2854	11.90
		Two-ACO	237.9	90.12	328.0	3.965	8.094	5.090	12.06
		Two-Greedy	274.9	93.54	368.4	4.582	9.730	6.613	14.31

distance to complete all tasks. It facilitates exploitation, but it is lack of exploration. It is easy to trap into a local optimal solution. Here, the greedy strategy for both the UAV and UGV is to select the nearest building as the next visiting building.

Genetic algorithm (Two-GA): it is a simple version of Two-MA without the local search. A comparison of Two-MA with Two-GA is made to demonstrate the superiority of the proposed MA framework over the classical GA framework and the effectiveness of the designed local search.

Two-ACO: It simulates the ant foraging process to execute the cooperative optimization for complex optimization problems. In Two-ACO, each ant is used to find a possible path for the time-aware path planning task, and the ants will converge a good path under the action of pheromone feedback.

B. Experimental Results on Synthetic Data Sets

In this part, all algorithms are tested on the four groups of synthetic data sets over 30 independent trials, and the statistical

results are recorded in Table VI. For each data set, we consider three different drone parameters of the UAV, and the optimal value is highlighted in boldface. In Table VI, the total distance is the sum of the UAV and UGV path distances, while the total time is the sum of the UAV and UGV path time and UGV. The UGV path time consists of the running time and the waiting time for charging the UAV.

From Table VI, we can see that **Two-MA** can obtain the path with a minimum of total execution time for all data. In other words, Two-MA enables the UAV/UGV cooperative system to complete the illegal building detection tasks with a minimum of execution time. This validates the superiority of Two-MA over Two-GA, Two-ACO and Two-greedy in finding an optimal path with a minimum of execution time.

Table VI also shows that Two-GA and Two-Greedy obtain the path with higher execution time than Two-MA and Two-ACO for all data sets, especially for the large-scale data set (Data index 4). This is to be expected that Two-GA is a lack of exploitation while Two-Greedy is a lack of exploration.

TABLE VII
RESULTS OF ALL ALGORITHMS ON THE REAL-WORLD DATA SETS

Setting	Index	Algorithms	Distance (km)			Time (hour)			
			UAV	UGV	Total	UAV	UGV	UGV-wait	Total
A	1	Two-GA	100.6	68.99	169.6	2.012	3.446	1.147	5.459
		Two-MA	96.52	74.83	171.4	1.930	2.929	0.4351	4.860
		Two-ACO	95.67	68.99	164.7	1.913	3.252	0.9527	5.166
		Two-Greedy	107.0	70.19	177.1	2.139	4.104	1.765	6.243
	2	Two-GA	144.0	122.9	267.0	2.881	6.219	2.121	9.100
		Two-MA	133.4	142.9	276.3	2.668	5.101	0.3365	7.769
		Two-ACO	135.0	122.7	257.7	2.700	6.037	1.947	8.737
		Two-Greedy	150.6	161.7	312.3	3.012	7.749	2.359	10.76
	3	Two-GA	250.2	207.8	457.9	5.003	12.46	5.536	17.46
		Two-MA	239.5	326.5	566.0	4.790	11.51	0.628	16.30
		Two-ACO	239.4	206.1	445.4	4.787	12.32	5.454	17.11
		Two-Greedy	266.5	211.9	478.3	5.329	14.19	7.129	19.52
	4	Two-GA	451.7	377.4	829.1	9.035	24.84	12.26	33.88
		Two-MA	423.4	616.8	1040	8.468	22.64	2.080	31.11
		Two-ACO	423.8	336.6	760.4	8.476	23.17	11.95	31.65
		Two-Greedy	476.7	523.4	1000	9.533	30.56	13.11	40.10
B	1	Two-GA	93.65	74.70	168.4	1.873	4.123	1.633	5.996
		Two-MA	87.96	83.01	171.0	1.759	3.428	0.6614	5.188
		Two-ACO	87.68	74.73	162.4	1.754	4.019	1.528	5.772
		Two-Greedy	97.63	76.23	173.9	1.953	4.933	2.392	6.886
	2	Two-GA	137.4	100.5	237.9	2.748	6.436	3.084	9.183
		Two-MA	127.9	134.5	262.4	2.558	5.625	1.141	8.183
		Two-ACO	127.3	100.1	227.4	2.546	5.944	2.608	8.490
		Two-Greedy	141.8	131.6	273.4	2.836	7.667	3.280	10.50
	3	Two-GA	251.2	226.8	478.0	5.024	12.57	5.008	17.59
		Two-MA	241.8	297.7	539.5	4.836	11.74	1.818	16.58
		Two-ACO	242.2	224.1	466.4	4.844	12.78	5.313	17.63
		Two-Greedy	274.4	254.6	529.0	5.487	15.64	7.150	21.12
	4	Two-GA	434.5	411.8	846.3	8.690	23.39	9.658	32.08
		Two-MA	417.0	499.8	916.8	8.341	21.13	4.476	29.48
		Two-ACO	416.9	364.9	781.8	8.338	22.20	10.04	30.54
		Two-Greedy	465.7	432.1	897.8	9.313	26.60	12.19	35.91
C	1	Two-GA	93.69	75.78	169.5	1.561	2.869	0.3427	4.430
		Two-MA	87.89	75.78	163.7	1.465	2.526	0	3.991
		Two-ACO	87.67	75.78	163.4	1.461	2.910	0.3840	4.371
		Two-Greedy	98.46	77.01	175.5	1.641	3.524	0.9565	5.165
	2	Two-GA	143.6	150.3	293.8	2.393	5.297	0.2887	7.690
		Two-MA	138.1	150.6	288.7	2.302	5.034	0.01305	7.335
		Two-ACO	137.9	153.1	291.1	2.299	5.755	0.6508	8.055
		Two-Greedy	156.6	165.4	322.0	2.610	7.096	1.582	9.706
	3	Two-GA	260.9	214.2	475.0	4.348	9.407	2.268	13.75
		Two-MA	250.9	245.6	496.5	4.182	8.608	0.4209	12.79
		Two-ACO	251.2	200.3	451.5	4.186	9.446	2.770	13.63
		Two-Greedy	287.4	216.8	504.2	4.790	10.89	3.666	15.68
	4	Two-GA	430.4	362.2	792.6	7.173	16.42	4.342	23.59
		Two-MA	410.1	414.0	824.2	6.835	14.89	1.086	21.72
		Two-ACO	409.2	325.5	734.7	6.820	15.47	4.620	22.29
		Two-Greedy	462.9	391.0	853.9	7.715	20.71	7.677	28.42

Those would result in that Two-GA hardly converges to a good solution in a given number of iterations while Two-Greedy easily gets trapped into a local optimal solution for the large-scale data set. Two-MA combines the GA and greedy optimizations to execute the exploration and exploitation, and therefore it can effectively find a good solution in a given number of iterations. Two-ACO uses the cooperative of ants and the pheromone feedback to execute the exploration and exploitation, respectively, and it also shows a good performance in finding an effective path for the UAV/UGV cooperation system to complete the illegal building detection tasks.

There is an interesting phenomenon in Table VI: Two-MA has a less task execution time than Two-ACO, whereas it has a longer path distance than Two-ACO. This is because detection of illegal tasks is modeled as a time-aware TSP problem, and Two-ACO neglects the cooperation schedules between the UAV and the UGV. Therefore, Two-ACO

takes a longer extra time than Two-MA for charging the UAV.

To further show the effectiveness of Two-MA, we show the planned path of Two-MA on the synthetic data in Fig. 9. The results show that Two-MA can effectively find good paths among buildings in both the same and different clusters.

Fig. 10 gives the averaged performance rankings of all algorithms on the simulated data. The results show that Two-MA has the smallest ranking value (1.167). This further validates the superiority of Two-MA over other baseline methods in planning paths for the UAV/UGV cooperation systems.

C. Experiments Results on Real-World Data Sets

In this part, we test all algorithms on four real-world data sets, and the experimental results over 30 independent trials are recorded in Table VII. The best results are highlighted

TABLE VIII
RESULTS OF ALL ALGORITHMS ON THE REAL-WORLD DATA UNDER A SPECIAL CASE IN WHICH THE UAV CAN BE QUICKLY CHARGED

Setting	Data index	Algorithms	Distance(km)			Time(hour)			
			UAV	UGV	Total	UAV	UGV	UGV-wait	Total
charge 5min	1	Two-GA	100.5	110.5	211.1	2.011	3.685	0	5.696
		Two-MA	94.89	110.5	205.4	1.898	3.685	0	5.583
		Two-ACO	94.66	110.5	205.2	1.893	3.685	0	5.578
		Two-Greedy	103.8	114.1	217.9	2.076	3.802	0	5.878
	2	Two-GA	144.3	158.7	303.0	2.886	5.289	0	8.175
		Two-MA	139.4	158.5	297.9	2.788	5.282	0	8.070
		Two-ACO	139.0	160.3	299.2	2.779	5.342	0	8.121
		Two-Greedy	154.0	173.4	327.4	3.080	5.779	0	8.859
	3	Two-GA	260.4	245.8	506.2	5.209	8.193	0	13.40
		Two-MA	247.9	232.9	480.8	4.957	7.764	0	12.72
		Two-ACO	247.7	248.9	496.5	4.953	8.296	0	13.25
		Two-Greedy	280.0	261.9	541.9	5.601	8.728	0	14.33
	4	Two-GA	453.3	441.4	894.7	9.066	14.71	0	23.78
		Two-MA	431.4	346.0	777.4	8.628	11.53	0	20.16
		Two-ACO	431.7	415.8	847.4	8.633	13.86	0	22.49
		Two-Greedy	482.8	457.6	940.4	9.656	15.25	0	24.91

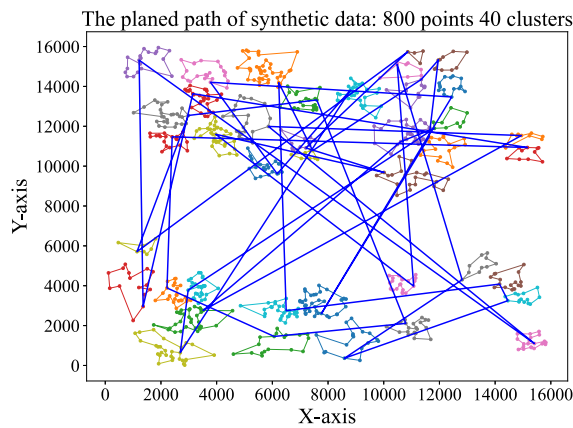


Fig. 9. Path of UAV and UGV planned by Two-MA in the synthetic data: 800 points. The blue lines represent the path of the UGV while the other color lines denote the path of the UAV.

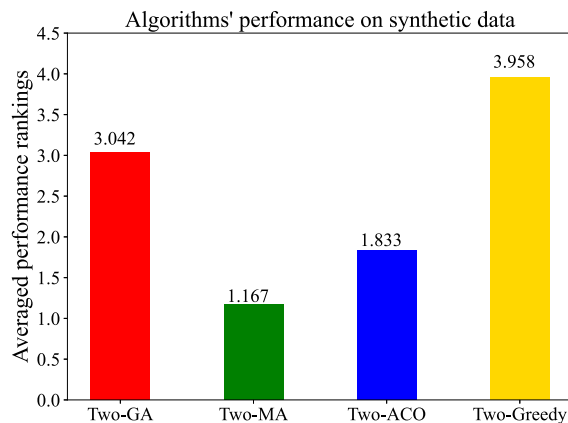


Fig. 10. Averaged performance rankings of all algorithms on the simulated data.

in boldface for each real-world data set. From Table VII, we can see that the total consumption time of Two-MA is shorter than that of Two-GA, Two-ACO, and Two-Greedy. This comparison validates the superiority of Two-MA over other comparison algorithms in completing the detection of illegal building in cities with a minimum task execution time.

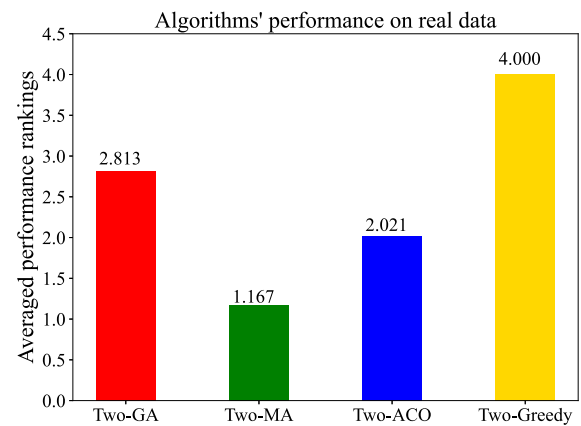


Fig. 11. Averaged performance rankings of all algorithms on the real-world data sets.

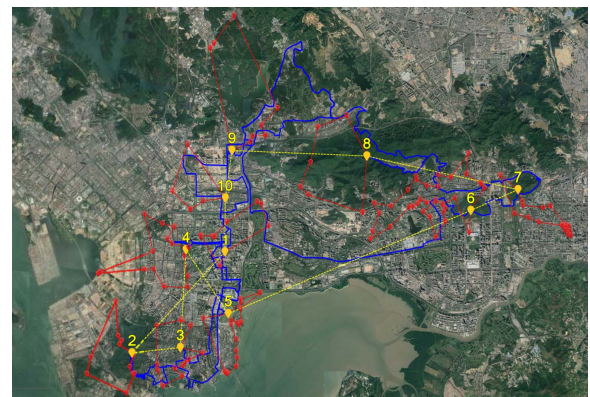


Fig. 12. Planned paths obtained by Two-MA for all parks in Nanshan and Futian District. The blue lines represent the path of the UGV while the red lines denote the path of the UAV. The yellow lines denote the visiting order of the UGV across different clusters.

Table VII shows some similar results as Table VI. Two-GA and Two-Greedy always have a poor performance, even for the small-scale real-world data set with 92 points and 6 clusters. Two-ACO shows a good performance in finding the shortest paths for the UGV whereas it has a poor performance in the task execution time. With the cooperative path planning of the UAV and UGV, Two-MA can effectively find the path

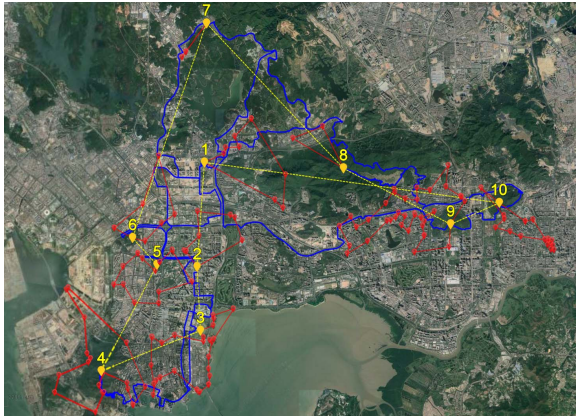


Fig. 13. Planned paths obtained by Two-MA for all parks in Nanshan and Futian District under the special case in which the UAV can be quickly charged. The blue lines represent the path of the UGV while the red lines denote the path of the UAV. The yellow lines denote the visiting order of the UGV across different clusters.

with a minimum task execution time for completing all illegal building detection tasks.

Table VII also illustrates that Two-MA has a father distance value than Two-ACO. This is because Two-MA generally plans a farther path for the UGV to charge the UAV, reducing the waiting time of the cooperative UAV/UGV system for completing all detection tasks.

Fig. 11 gives the averaged performance rankings of all algorithms on the real-world data sets. The results show that Two-MA has the smallest ranking value (1.167). This further validates the superiority of Two-MA in planning paths for the UAV/UGV cooperation systems.

Fig. 12 shows the path planning of Two-MA for the UAV and UGV cooperative system on the real data with all parks in Nanshan and Futian District of Shenzhen City. The path length for the UAV inside the 10 clusters is 133.38 km, while the path length for the UGV is 142.94 km. In this case, we can use the UAV to detect nonstandard buildings in the same clusters, and adopt the UGV to take the UAV across different clusters. The task execution time for the UAV inside the 10 clusters is 2.67 h, while the execution time for the UGV is 5.10 h. Moreover, the extra waiting time for the charging of the UAV is 0.34 h.

To further show the effectiveness of Two-MA, we test all algorithms under a special case in which the UAV can be quickly charged, and record the results on the real-world data sets into Table VIII. In this case, the time-aware TSP optimization problem can be considered as a path-aware TSP problem. The results show that Two-MA finds the shortest paths for the UGV while Two-ACO searches the shortest paths for the UAV. This validates the effectiveness of both Two-MA and Two-ACO in solving the path-aware TSP optimization problem.

Fig. 13 shows the path planning of Two-MA for the UAV and UGV cooperative system on the real data with all parks in Nanshan and Futian District of Shenzhen City under the special case. The result shows that there are no obvious intersection points of the planned paths for both the UAV and UGV. This further validates the effectiveness of the proposed Two-MA in solving the path-aware TSP optimization problem.

V. CONCLUSION

In this article, we proposed a UAV/UGV cooperative detection system for the detection of nonstandard buildings in cities, in which the UAV and UGV operate alternately under the consideration of UAV's energy and UGV's loading capacity. Moreover, we proposed a two-level memetic path planning algorithm (Two-MA) to find the paths with the least task execution time from both the node and cluster levels. In addition, experimental results on the synthetic data as well as the real data validated the superiority of Two-MA over the classical algorithms (Two-GA, Two-ACO, and Two-greedy) in finding the paths with minimum task execution time for the detection of nonstandard buildings in cities.

In future work, we will study the multiple UAVs/UGVs cooperative detection system for the detection of nonstandard buildings in cities under more real constraints. Moreover, we will take different terrain conditions into account while assigning tasks to different vehicles for the whole cooperative system. In addition, we will devise a multitasking evolutionary algorithm to cooperatively plan paths for multiple UAVs/UGVs. Finally, we will propose a parallel algorithm for solving large-scale path planning problems in the UAV/UGV cooperative system.

ACKNOWLEDGMENT

Part of the results appeared in IEEE ARM 2020 [1].

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